

AI/ML Internship – Day 52 Notes & Tasks

Module 7: Text Preprocessing – Tokenization, Stop Words & Text Cleaning

Part 1: Recap of Day 51

Yesterday we learned:

-  What is NLP
-  Applications of NLP
-  Text Data
-  NLP Workflow
-  NLP vs Computer Vision

Today we will learn the first practical step in NLP:

 Text Preprocessing

Part 2: What is Text Preprocessing?

Definition

Text Preprocessing is the process of converting raw text into a clean format that computers can understand.

Before processing:

Hello!!! How are you???? 😊😊

After preprocessing:

hello how are you

Why Do We Need Text Preprocessing?

Real-world text contains:

- Emojis
- Symbols
- Punctuation

- Special Characters
- Unnecessary Words

Computers cannot easily understand noisy text.

Therefore we clean the text before analysis.

Part 3: NLP Pipeline

Raw Text



Text Cleaning



Tokenization



Stop Word Removal



Feature Extraction



Model Training

Part 4: What is Tokenization?

Definition

Tokenization is the process of breaking text into smaller units called tokens.

Tokens can be:

- Words
 - Sentences
 - Characters
-

Example

Text:

I love AI

Tokens:

["I", "love", "AI"]

Visual Representation

I love AI

↓

["I", "love", "AI"]

Part 5: Types of Tokenization

1. Word Tokenization

Text:

AI is amazing

Output:

["AI", "is", "amazing"]

2. Sentence Tokenization

Text:

AI is amazing.

Python is easy.

Output:

```
[  
  "AI is amazing.",  
  "Python is easy."  
]
```

3. Character Tokenization

Text:

AI

Output:

['A', 'I']

Part 6: Tokenization Using Python

```
text = "I love AI"
```

```
tokens = text.split()
```

```
print(tokens)
```

Output:

```
['I', 'love', 'AI']
```

Part 7: What are Stop Words?

Definition

Stop Words are commonly used words that usually do not add important meaning.

Examples:

is
am
are
was
were
the
a
an
of
in
on
at

Example

Sentence:

I am learning AI

After removing stop words:

learning AI

Part 8: Why Remove Stop Words?

Suppose we have:

The dog is running in the park

Important words:

dog

running

park

Less important:

the

is

in

the

Removing stop words helps the model focus on meaningful words.

Part 9: Common English Stop Words

a

an

the

is

am

are

was

were

of

to

for

and

or

in

on

at

with

from

Part 10: Stop Word Removal Example

Sentence:

The boy is playing football in the ground

Output:

boy
playing
football
ground

Part 11: What is Text Cleaning?

Definition

Text Cleaning means removing unwanted content from text.

Remove Punctuation

Input:

Hello!!!

Output:

Hello

Remove Numbers

Input:

AI 2025

Output:

AI

Remove Special Characters

Input:

Hello@#\$%

Output:

Hello

Convert to Lowercase

Input:

HELLO

Output:

hello

Part 12: Lowercase Conversion

Why?

Computer treats:

AI

and

ai

as different words.

Convert everything to lowercase:

```
text = "AI IS AWESOME"
```

```
print(text.lower())
```

Output:

ai is awesome

Part 13: Remove Punctuation

```
import string
```

```
text = "Hello!!!"
```

```
clean_text = text.translate(  
    str.maketrans("", "", string.punctuation)  
)
```

```
print(clean_text)
```

Output:

Hello

📌 Part 14: Remove Numbers

```
import re

text = "AI 2025"

clean_text = re.sub(r'\d+', "", text)

print(clean_text)
```

Output:

AI

📌 Part 15: Complete Text Cleaning Example

```
import re

text = "Hello!!! AI 2025 is Amazing."

text = text.lower()

text = re.sub(r'[^a-zA-Z\s]', "", text)

print(text)
```

Output:

hello ai is amazing

📌 Part 16: Real-World Example

Amazon Review

Original:

This product is AWESOME!!! 🥰🥰🥰

After Cleaning:

this product is awesome

After Stop Word Removal:

product awesome

After Tokenization:

["product", "awesome"]

Part 17: Why Preprocessing is Important?

Without preprocessing:

-  Poor accuracy
 -  Duplicate words
 -  Noise in data
-

With preprocessing:

-  Clean data
 -  Better accuracy
 -  Faster training
 -  Better NLP models
-

Part 18: Theory Questions

What is Text Preprocessing?

Cleaning and preparing text for NLP.

What is Tokenization?

Breaking text into smaller units called tokens.

What are Stop Words?

Common words that add little meaning.

Why remove stop words?

To focus on meaningful information.

What is Text Cleaning?

Removing unwanted characters and noise from text.

Part 19: Interview Questions

Why is text preprocessing important?

It improves data quality and model performance.

What is a token?

A word, sentence, or character obtained after tokenization.

Give examples of stop words.

is, am, are, the, a, an.

Why convert text to lowercase?

To maintain consistency.

What is the first step in NLP?

Text preprocessing.

Tasks for Day 52

Theory

1. Define Text Preprocessing.
2. What is Tokenization?
3. What are Stop Words?
4. Why remove punctuation?
5. Why convert text to lowercase?

Practical Task 1

Tokenize:

I love Machine Learning

Practical Task 2

Convert:

HELLO WORLD

to lowercase.

Practical Task 3

Remove punctuation from:

Python!!! is Awesome???

Practical Task 4

Identify stop words in:

The cat is sitting on the chair

Practical Task 5

Clean this sentence:

AI 2025 is AMAZING!!! 😍